

Requirements Traceability Matrix (RTM)

IBA Campus Facility Booking System

By:

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1 Introduction

The Requirements Traceability Matrix (RTM) is a sub-deliverable of the Test Plan for the IBA Campus Facility Booking System. Its purpose is to ensure that all functional requirements defined in the Software Requirements Specification (SRS) are mapped to specific test cases. This ensures 100% test coverage and validates that every system feature is verified during the testing lifecycle.

2 Traceability Matrix

Table 1: Mapping Functional Requirements to Test Cases

Req. ID	Requirement Description	Priority	Mapped Test Case ID(s)
2.1 Authentication (Student / PO)			
REQ-1	Login form for students (ERP/Pass)	High	TC-AUTH-001
REQ-2	Login form for PO (User/Pass)	High	TC-AUTH-002
REQ-3	Validate credentials against data	High	TC-AUTH-001, TC-AUTH-002
REQ-4	Redirect to respective dashboard	High	TC-AUTH-001, TC-AUTH-002
REQ-5	Error message for invalid login	High	TC-AUTH-003
REQ-6	Prevent unauthorized dashboard access	High	TC-AUTH-004, TC-AUTH-005, TC-AUTH-006
2.2 Request Room Booking (Student)			
REQ-1	Display list of buildings	High	TC-BOOK-001
REQ-2	Display available rooms in building	High	TC-BOOK-001
REQ-3	Select date/time (9AM-6PM)	High	TC-BOOK-004, TC-BOOK-005, TC-BOOK-006, TC-BOOK-007

Table 1 – Continued from previous page

Req. ID	Requirement Description	Priority	Mapped ID(s)	Test Case
REQ-4	Prevent booking if room is busy/disabled	High	TC-BOOK-002, TC-BOOK-003	TC-
REQ-5	Allow entry of booking purpose	Medium	TC-BOOK-008	
REQ-6	Forward request to PO & Confirm	High	TC-BOOK-001	
REQ-7	Show conflict error message	High	TC-BOOK-002, TC-BOOK-009	TC-
REQ-8	Maintain booking record	Medium	TC-DATA-001	
2.3 Accept/Reject Room Booking (PO)				
REQ-1	List all pending requests	High	TC-PO-001	
REQ-2	Allow PO to approve or reject	High	TC-PO-002, TC-PO-003	TC-PO-
REQ-3	Update room schedule if approved	High	TC-PO-002	
REQ-4	Notify student of decision	High	TC-PO-002, TC-PO-003	TC-PO-
REQ-5	Prevent double booking if approved	High	TC-PO-004	
2.4 & 2.5 Add Student / PO Member (Admin)				
REQ-1	Accept user details (ERP, Name, etc.)	High	TC-ADMIN-001, TC-ADMIN-003	TC-
REQ-2	Validate unique ERP/Username	High	TC-ADMIN-002	
REQ-3	Create account and notify success	High	TC-ADMIN-001, TC-ADMIN-003	TC-
REQ-4	Error on invalid input or exists	Medium	TC-ADMIN-002	
2.6 Disable Room (Admin)				
REQ-1	Admin select and disable room	Medium	TC-ADMIN-006	
REQ-2	Prevent future bookings for disabled	Medium	TC-BOOK-003, TC-ADMIN-006	TC-
REQ-3	Show "disabled" status	Medium	TC-ADMIN-006, TC-ADMIN-007	TC-
2.7 & 2.8 Add Room / Building (Admin)				
REQ-1	Accept Room/Building details	Medium	TC-ADMIN-004, TC-ADMIN-005	TC-
REQ-2	Validate unique names	Medium	TC-ADMIN-004, TC-ADMIN-005	TC-
REQ-3	Add to database	Medium	TC-ADMIN-004, TC-ADMIN-005	TC-
REQ-4	Show confirmation message	Medium	TC-ADMIN-004, TC-ADMIN-005	TC-
2.9 Cancel Booking (Student/PO/Admin)				

Table 1 – Continued from previous page

Req. ID	Requirement Description	Priority	Mapped ID(s)	Test Case
REQ-1	Allow cancellation before start time	Medium	TC-CANCEL-001, TC-CANCEL-002, TC-CANCEL-003, TC-CANCEL-004	
REQ-2	Update status to "Cancelled"	Medium	TC-CANCEL-002	
REQ-3	Free the room slot after cancel	Medium	TC-CANCEL-002, TC-CANCEL-005	

3 Summary of Coverage

- **Total Functional Requirements:** 37
- **Total Test Cases Defined:** 33
- **Test Coverage:** 100% (Every requirement is mapped to at least one test case).
- **High Priority Coverage:** All "High" priority requirements are covered by multiple functional and negative test scenarios.